Group video and presentation instructions for Télécom Paris AI Ethics classes 2025-2026

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Instructions: This exercise is to simulate an actual discussion that might occur in a company or administration between different stakeholders regarding an AI project. This is to test your ability to put yourself in the shoes of a particular person in the company whose job is to defend certain priorities (operational, financial, ethical, compliance, safety), and reach a recommendation to present to management, such as “this project is raises concerns and priorities that can conflict. We will present you those concerns and priorities and then propose a way to reconcile them”

Please form groups of 4-5 and each person assumes a role as described in the “role cards”. Each group chooses a case study and becomes expert on the case study by reading relevant documents. Then you produce a 7 minute video (see video instructions at the end of this document).

During the class presentations, we will project the video, then a jury will pretend to be the executive board of the company and ask you questions. It will go fast, and members of the jury will challenge your conclusions, like in real life when you present something to senior management.

You are free to choose another use case as long as you talk to us about it first.

Once you have studied the case and have prepared your different positions, you are also expected to contact one of us for a coaching session (30 min) in person or by zoom. If you don’t seek out prior coaching from one of us the final presentation will suffer.

Roles cards

Computer scientist
Your work as a computer scientist led you to having a deep technical knowledge of the models deployed for the use case. Your goal is to make sure that the proposed solution is technically feasible in the allotted time with respect to the state of the art in machine learning and artificial intelligence.

Regulation / compliance officer
You are working as a compliance officer for the AI Act. Your goal is to ensure that the use case is compliant with the EU AI Act.
Your role is to assess whether the system considered is high-risk or not, and whether the solutions proposed by the company are adequate with respect to the identified level of risk.

(AI) Ethics officer
You are working as a social scientist on the ethics of AI, and have a strong knowledge of biases and inequalities and how they are amplified to the prism of current AI. Your goal is to ensure the proposal for the use case does not amplify inequalities or contribute to asymmetric power structures.

Cybersecurity expert
You are an expert in cybersecurity and information systems security. Your goal is to ensure that the organization's IT systems meet requirements of safety, integrity, confidentiality and availability.

Project manager
Your role is to be able to discuss and challenge every other member of the committee, and ensure the cohesiveness of the proposed recommendation for the case study.

Case studies

ETIAS case

You are on a team within the French government asked to develop ML solutions to help automated border control pursuant to the European Union's ETIAS system. Please present to the prime minister's team the proposed solutions and the different opinions of stakeholders in your group, including trade-offs that may be needed to reconcile the different viewpoints.

Some reading to get you started:

Jo Pesch, P., Dimitrova, D., Boehm, F. (2022). Data Protection and Machine-Learning-Supported Decision-Making at the EU Border: ETIAS Profiling Under Scrutiny. In: Gryszczyńska, A., Polański, P., Gruschka, N., Rannenberg, K., Adamczyk, M. (eds) Privacy Technologies and Policy. APF 2022. Lecture Notes in Computer Science(), vol 13279. Springer, Cham. https://doi.org/10.1007/978-3-031-07315-1_4

https://edri.org/our-work/protect-not-surveil-eu-ai-act-fails-migrants-people-on-the-move/

https://home-affairs.ec.europa.eu/policies/schengen-borders-and-visa/smart-borders/european-travel-information-authorisation-system_en

https://www.europarl.europa.eu/RegData/etudes/IDAN/2021/690706/EPRS_IDA(2021)690706_EN.pdf

Content moderation on large online platforms

You work for Meta. You have been asked to improve the current AI system for detecting and removing content that violates Meta's "bullying and harassment" policy, and in particular content that is sexually degrading. The current system has a high false negative rate, i.e. it misses too many instances of sexually harassing content. Also, the error rate is much higher for non-English languages, and for certain population groups, creating a problem of bias. Meta management wants you to see if LLMs might improve the current system. Please present recommendations pointing out the friction points and risks raised by the different stakeholders in your group.

Some reading to get you started (but don't limit yourself to this!):

https://arxiv.org/abs/2409.03219

https://ojs.aaai.org/index.php/ICWSM/article/view/31358

https://www.ofcom.org.uk/siteassets/resources/documents/research-and-data/online-research/online-harms/2023/content-moderation-report.pdf?v=330128

Algorithmic Arbitrariness in Content Moderation
2024 https://facctconference.org/static/papers24/facct24-151.pdf

Recommender systems and democratic processes

A very large online platform offers feeds of posts (news updates, articles, comments by other users) to its users. As such, the content — and its order — on each feed is determined by an AI system using user features, post features, post engagement, past interactions, etc. The field of recommender systems, initially spearheaded by music and movie recommandation — two relatively low-stakes applications — now raises significant ethical and societal concerns.

References

- The DMA and DSA explained: https://www.europarl.europa.eu/topics/en/article/20211209STO19124/eu-digital-markets-act-and-digital-services-act-explained
- A research article on filter bubbles and echo chambers : https://academic.oup.com/poq/article/80/S1/298/2223402
- A research article questioning the existence of filter bubbles : https://wires.onlinelibrary.wiley.com/doi/full/10.1002/widm.1512?casa_token=vmFEZXnOQnoAAAAA%3AQm6ypu_OIBiFX-KYQiFNz9qOCytB9sZK-WTLl9nTpf_-RwYRooNuTyCE3m1pt-4cionlUAPrprGVmQ

Automatic videosurveillance/facial recognition during a large-scale event

A government agency in France has implemented an AI-based system to automatically identify certain events (nominally, terrorism risk and stampedes) automatically based on the video flux of people in a given area. This raises multiple ethical, social and legal questions, based on the legal regime under which this system operates, the area and duration in time, the notification to passer-bys, etc.

References

- Report from la Quadrature du Net on facial recognition at the Paris Olympics : https://www.laquadrature.net/2024/07/30/jeux-olympiques-fichage-de-masse-et-discrimination-politique/
- EPDB report on facial recognition in the EU : https://www.edpb.europa.eu/news/news/2023/edpb-adopts-final-version-guidelines-facial-recognition-technology-area-law_en
- The bias in facial recognition (US article) : https://www.aclu-mn.org/en/news/biased-technology-automated-discrimination-facial-recognition
- Resarch article on public attitudes towards facial recognition : https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8513835/

Automatic generation of fraud risk scores for recipients of social benefits (social allowances) to help human agents prioritise fraud investigations

A government agency has implemented an AI-based system to automatically generate fraud risk scores for recipients of social benefits. This tool is designed to help human agents prioritize fraud investigations by flagging households who are deemed high-risk based on data held by public institutions like tax and financial history, demographic data, and behavioral patterns. While the system aims to improve efficiency and reduce fraud, several ethical and legal concerns have emerged.

- Quadrature du Net reveals French CAF’s algorithm : https://www.laquadrature.net/en/2023/11/27/scoring-of-welfare-beneficiaries-the-indecency-of-cafs-algorithm-now-undeniable/

- Defenseur des droits and CNIL's report on preventing automated discrimination : https://www.defenseurdesdroits.fr/sites/default/files/2023-07/ddd_rapport_algorithmes_2020_EN_20200531.pdf
- Le Monde's investigation on CAF's algorithmic practices : https://www.lemonde.fr/en/les-decodeurs/visuel/2023/12/05/how-an-algorithm-decides-which-french-households-to-audit-for-benefit-fraud_6313254_8.html

A specially-trained LLM to advise consumers on their legal rights and help them draft claims (a robo-lawyer).

A tech company has developed a user-facing app based on a specially-trained large language model (LLM), called "Robo-Lawyer," to provide consumers with legal advice and assist in drafting claims. It is marketed as a cost-effective solution to address access to justice, particularly for those unable to afford traditional legal services. The Robo-Lawyer uses AI to analyze user inputs, suggest legal options, and generate documents for court proceedings. However, concerns have arisen about the ethical implications of this tool.

Useful resources:

- Stanford study – Hallucinating Law: Legal Mistakes with LLM are Pervasive – Jan 2024 : https://law.stanford.edu/2024/01/11/hallucinating-law-legal-mistakes-with-large-language-models-are-pervasive/
- Stanford research paper – Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools – May 2024 : https://dho.stanford.edu/wp-content/uploads/Legal_RAG_Hallucinations.pdf
- On the use of LLM to adjudicate complex cases : https://adamunikowsky.substack.com/p/in-ai-we-trust ; https://adamunikowsky.substack.com/p/in-ai-we-trust-part-ii
- 2023 Year-End Report on the US Federal Judiciary, highlighting the risks of AI in courts : https://www.supremecourt.gov/publicinfo/year-end/2023year-endreport.pdf
- US lawyer cited fake cases in court : https://www.forbes.com/sites/mollybohannon/2023/06/08/lawyer-used-chatgpt-in-court-and-cited-fake-cases-a-judge-is-considering-sanctions/

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LLM-as-a-service platform in banking services

A banking company is developing an internal LLM-as-a-service platform for all its employees. The company chooses LLMs, whether open-source or proprietary, and makes them available for use by its employees from all services. Employees can then build applications based on the chosen LLM for their own needs. This can lead to various applications like customer-facing chatbots, credit scoring algorithms for granting loans to individuals, and credit card or financial

fraud detection. The bank thinks that some AI use cases in banking can lead to ethical and legal risks, it wonders how to ensure AI applications are built and deployed by its teams while ensuring trustworthiness and legal compliance. The bank also wonders what organization and processes it should put in place to ensure that employees using the LLM-as-a-service platform develop and deploy AI applications ethically.

Useful resources:

https://cdn.ceps.eu/wp-content/uploads/2025/05/Unfit-for-purpose-ECRI-In-Depth-Analysis.pdf

Ali, A., Abd Razak, S., Othman, S. H., Eisa, T. A. E., Al-Dhaqm, A., Nasser, M., Elhassan, T., Elshafie, H., & Saif, A. (2022). Financial Fraud Detection Based on Machine Learning: A Systematic Literature Review. *Applied Sciences*, *12*(19), 9637. https://doi.org/10.3390/app12199637

J. O. Awoyemi, A. O. Adetunmbi and S. A. Oluwadare, "Credit card fraud detection using machine learning techniques: A comparative analysis," *2017 International Conference on Computing Networking and Informatics (ICCNI)*, Lagos, Nigeria, 2017, pp. 1-9, doi: 10.1109/ICCNI.2017.8123782.

https://www.solutions-numeriques.com/cas-dusage-dia-generative-bnp-paribas-met-en-place-en-interne-une-plateforme-de-llm-as-a-service/

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Dienst Uitvoering Onderwijs (DUO) system – The Netherlands

DUO is a rule-based risk profiling system used to detect the abuse of out-of-home student grants. The system is overseen by DUO caseworkers, who prioritized flagged cases for investigation based on risk score and manual selection and exclusion criteria. Privately contracted inspectors then carried out short but invasive investigations of the selected students' residences and – based on their intuitions about how a student's living environment should look – reported whether they believed there was fraud.

DUO has been shut criticized because students with a non-European migration background were behind most of the legal cases brought against the system. Racialized groups and students with lower levels of education were disproportionately flagged and subjected to invasive home investigations.

Your team is hired by the Dutch government to understand what is going wrong with it and to propose appropriate measures to solve the issue.

- Amnesty International

- Algorithm Audit (+ addendum)

Video guidelines

Video duration: 7 minutes

The goal of the video is to give us a concise overview of your discussions and compromises before the final exam. We expect you to explain the consensus you have reached on your case study, and how compromises were made when competing interests surfaced. It is expected (and valued) that you will discuss the problems you encountered while negotiating a solution of your case study.

The video can be done facing camera or as a voice-over, or a mixture of both. Every group member should talk at some point in the video. The final video must not exceed 7 minutes.

The jury will already be aware of the case study, you do not need to spend time explaining it. You can give a brief overview if you wish to. A possible plan is to give an overview of the case study and your solution (1mn), spend 1mn per group member to explain their role, the trade-offs made and how they were made (1mn/group member), and 1mn summarising your recommendation on the case study.

Your final recommendation can be anything as long as it is backed up with good arguments. It can lead to not using AI/machine learning, to using it within constraints, etc. We will focus on the quality of your arguments rather than the contents of the recommendation itself.